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RAG-Based Legal Document Analyzer

1. Work Completed So Far

The project currently includes a React-based frontend and a Python FastAPI backend. The implemented features include:

- **PDF Upload and Processing:** Upload, validate, and extract text from legal PDF documents using PyMuPDF.
- **Document Management:** View uploaded documents, check their status, and delete documents.
- **RAG-Based Question Answering:** Retrieve relevant document content using embeddings and a FAISS vector store to answer user questions.
- **Source References:** Display relevant text snippets and page numbers alongside answers.
- **LLM Integration:** Support for OpenAI and Google Gemini, with a basic offline answering mode.
- **User Interface:** Dashboard, document management page, and chat interface.

2. Current Project Progress

Approximately **50–55% of the planned project** has been implemented. The core workflow of uploading a PDF, extracting its text, indexing the content, and asking document-specific questions is functional.

3. Future Work

The remaining work will focus on:

- Adding OCR support for scanned PDFs.
- Integrating PostgreSQL for persistent document storage.
- Implementing document summarization and clause identification.
- Adding multi-document question answering and chat history.
- Implementing user authentication.
- Testing, improving reliability, and deploying the application.

4. Expected Outcome

The final system will provide a web-based platform for analyzing legal documents, answering questions using their contents, identifying important clauses, and generating summaries to support preliminary document review.